Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

Headline Finding

50% task-completion time horizon doubles every ~207 days (~7 months)

Source

METR (Kwa, West, Becker, et al.)

arXiv:2503.14499 · March 2025

Current Frontier

o3 ≈ 1 hour 50 minutes

vs GPT-2 ≈ 2 seconds (2019)

A New Metric: 50% Task-Completion Time Horizon

Duration where model succeeds on 50% of tasks (human-time calibrated)

1) Task Suite
170 tasks across software/resear

2) Measurement
Human timing + AI success rates

3) Logistic Fit
Estimate 50% success duration

Methodology figure from source:

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Figure 2: Benchmark methodology overview (task suite + human baselines + logistic modeling)

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effo

Suite composition + example task duration spectrum

800+ human baselines · 2,529 hours

Task-Suite Sources

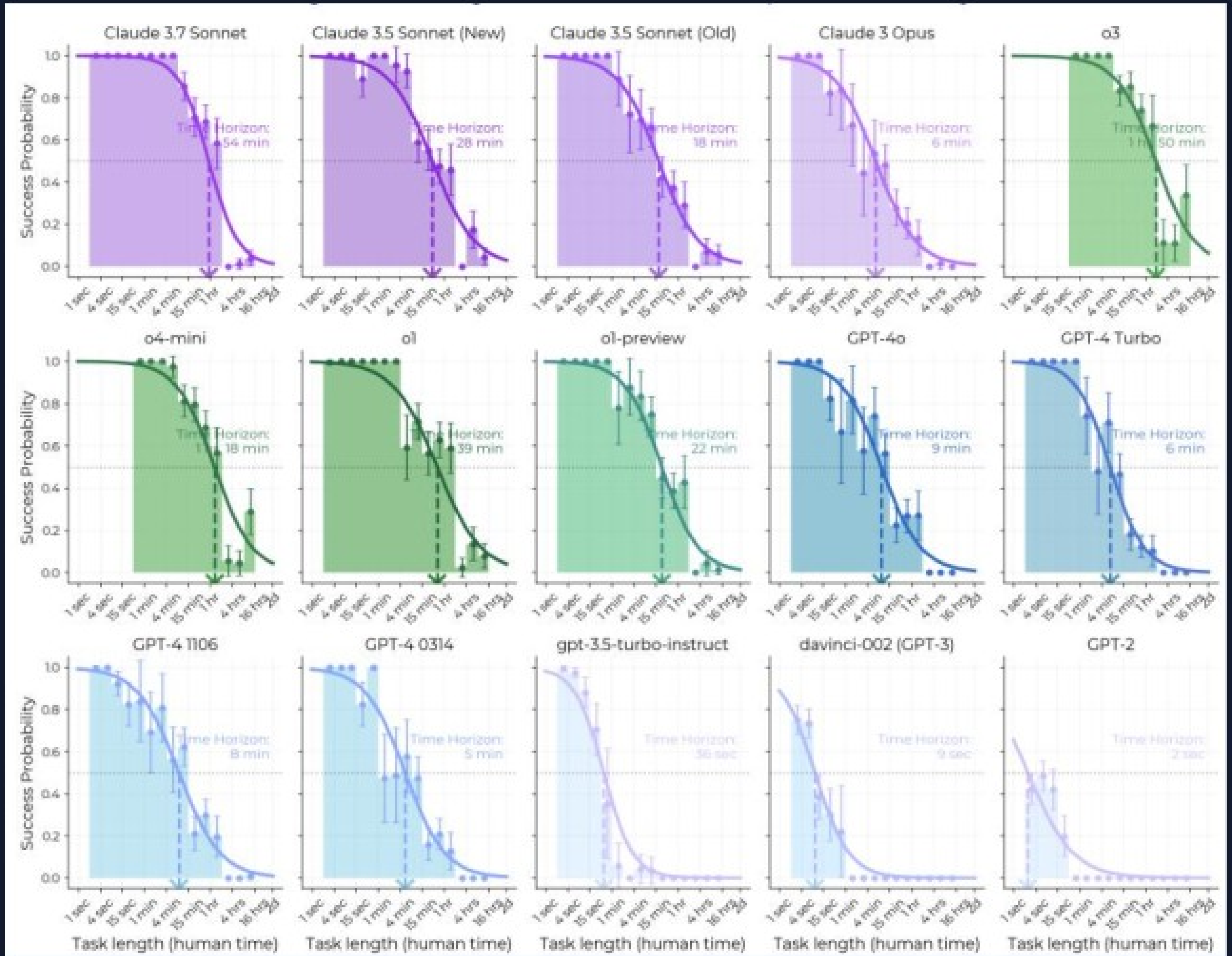
Source	Task Count	Human Time	Example Scope
HCAST	97	1 min–30 hrs	Diverse software engineering tasks
RE-Bench	7	8 hrs	Hard ML research engineering
SWAA	66	1 sec–30 sec	Software atomic actions

Example Tasks Across the Time Range

Family	Human Time	Description
find_shell_script	3 seconds	Identify which file is a shell script
wikipedia_research	1 minute	Research simple factual information
oxdna_simple	9 minutes	Fix bug in molecular dynamics inputs
munge_data	56 minutes	Write JSON transformation script
cuda_backtesting	8 hours	Implement custom CUDA kernels (30x speedup)

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 M

Exponential trend in 50% task-completion horizon by model release date



Doubling Time

~207 days

(~7 months)

$R^2 = 0.97$

Milestones

GPT-2 (2019): ~2 sec

GPT-4 0314 (2023): ~5 min

o3 (2025): ~1h 50m

Trend measured across 2019–2025 model rele

Success Probability Falls with Task Length for All Models

Frontier models shift curves rightward, extending autonomous task duration

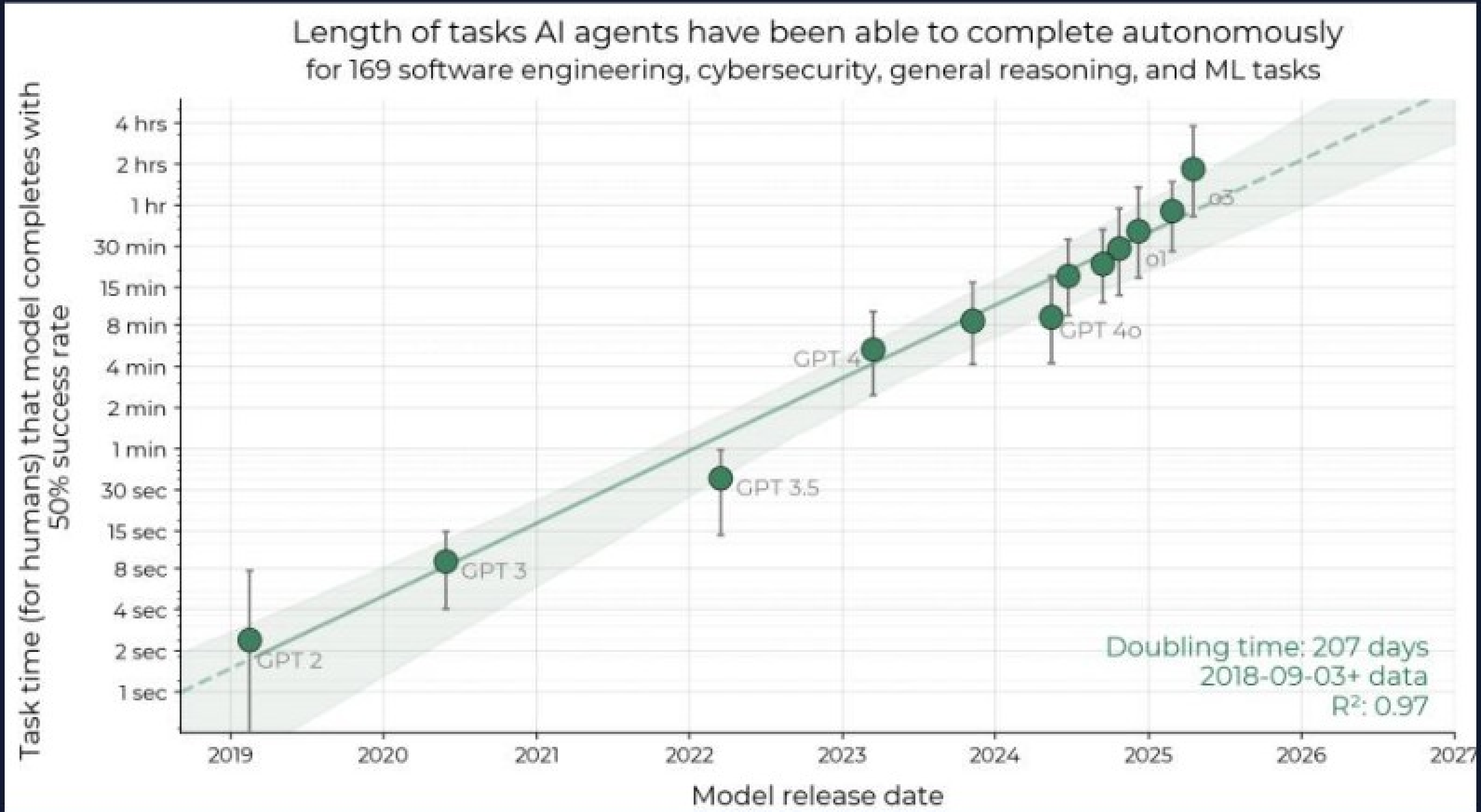


Figure 3: Per-model success probability curves versus task length

Frontier Shift

Claude 3.7 Sonnet:
~54 minutes
o3: ~1 hour 50 minutes

Baseline Contrast

GPT-2 horizon: ~2 seconds

Large multi-order-of-magnitude gap

Additional Signal

80% horizon follows similar trend but ~5x shorter

Reasoning, Tool Use, and Error Recovery Drive Growth

Capability advances increase success on longer multi-step software tasks

1. Improved Logical Reasoning

Better decomposition and chain-of-thought plannin

2. Better Tool Use

More effective code, shell, and workflow tool executio

3. Greater Reliability

Improved self-awareness and consistency under co

4. Adaptation to Mistakes

Recovers from errors and continues toward completio

Limitation: Performance drops on less structured, messier tasks

External validity remains open: benchmark tasks may not perfectly represent real-world intellectual labor distributions, despite supplementary checks.

Extrapolation: Month-Long Task Automation by 2028–20

If current trend persists, autonomous horizon reaches 1 month (167 work hours)

Forecast Window: mid-2028 to mid-2031

Target horizon: 167 work hours (~1 month)

Naive extrapolation from observed 2019–2025 exponential fit

Why This Could Arrive Sooner

- 2023–2025 growth ≈20% faster than overall 2019–2025 rate
- Frontier systems improving on reasoning + tool orchestration

Critical Caveats

- External validity is uncertain
- Real-world task distribution differs
- Trend may bend with bottlenecks, regulation, or eval changes

Takeaways: Exponential Capability Growth Requires Proact

1. 50% time horizon is a robust, intuitive autonomy metric ($R^2 = 0.97$ over 6 years).
2. Growth pace is rapid: doubling every ~7 months, with possible acceleration since 2024.
3. Current frontier (o3) reaches ~1 hour 50 minutes of skilled-human-equivalent work.
4. External validity remains open; benchmark tasks may not fully represent real-world labor.
5. Key limits persist on unstructured tasks and uncertain generalization across job domains.

Safety governance implication: tighten eval cadence, capability thresholds, and deployment controls ahea